

HVIDE SANDE, Denmark

WMO: 060580

Lat: 56.007N

Lon: 8.141E

Elev: 16

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	19.4	23.2	15.1	11.8	21.0	18.9	14.2	25.2	38.8	38.0	34.6	37.1	13.6	60	0.768

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.0	76.4	67.2	72.8	65.7	69.9	64.0	68.5	74.3	66.5	71.8	64.8	69.0	10.7	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	96.8	71.9	64.3	90.2	69.4	62.8	85.5	67.8	32.7	74.6	31.1	72.1	29.8	69.0	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.5	29.9	27.0	DB	17.7	81.6	7.5	4.0	12.2	84.5	7.8	86.8	3.6	89.1	-1.9	92.0	(4)
(5)			WB	16.7	71.0	7.1	2.6	11.6	72.9	7.4	74.4	3.4	75.8	-1.8	77.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.9	35.9	35.4	38.7	45.7	52.6	57.6	62.8	63.5	58.5	51.6	44.1	39.0
	DBStd	11.19	6.42	5.88	4.79	5.09	5.48	4.65	4.98	3.85	3.61	4.42	5.41	6.80
	HDD50	1934	436	409	349	150	29	1	0	0	0	36	183	341
	HDD65	5964	901	829	814	578	387	228	105	76	197	417	626	806
	CDD50	1524	0	0	0	22	110	229	397	418	256	84	8	0
	CDD65	79	0	0	0	0	2	7	38	29	3	0	0	0
	CDH74	209	0	0	0	0	5	17	122	63	2	0	0	0
	CDH80	19	0	0	0	0	0	1	13	5	0	0	0	0
Wind	WSAvg	14.6	16.5	15.4	14.8	13.6	13.6	13.9	12.7	13.5	14.6	15.6	16.0	15.5
Precipitation	PrecAvg	32.1	2.7	2.0	1.9	1.5	1.7	2.4	2.4	3.3	3.5	3.9	3.4	3.1
	PrecMax	40.1	4.5	3.7	4.0	2.7	3.0	5.6	3.9	5.8	6.7	8.2	6.9	6.9
	PrecMin	22.5	0.4	0.4	0.5	0.2	0.1	0.5	0.5	0.9	1.3	1.1	1.0	0.9
	PrecStd	4.8	1.1	0.9	1.0	0.6	0.8	1.4	0.9	1.3	1.5	1.9	1.3	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.4	46.2	51.5	64.3	73.4	76.3	82.0	80.1	72.1	62.1	56.4	50.8
		MCWB	46.2	43.9	46.6	55.5	63.3	64.9	69.8	69.3	65.6	59.3	54.2	48.8
	2%	DB	46.5	44.7	47.4	60.0	68.5	71.5	77.5	75.1	67.6	60.0	54.2	49.2
		MCWB	44.3	42.9	44.2	53.3	60.0	62.9	68.3	67.2	62.4	57.2	52.0	47.2
	5%	DB	45.1	43.5	45.5	56.6	64.8	67.6	73.8	71.7	65.3	58.5	52.6	48.0
		MCWB	43.2	41.9	43.1	51.2	57.5	61.6	66.6	65.5	60.9	55.9	50.4	46.3
	10%	DB	43.9	42.5	44.1	53.3	61.0	64.4	70.2	69.0	63.5	57.3	51.0	46.9
		MCWB	42.1	41.1	42.2	49.0	55.6	59.8	64.7	64.1	59.6	54.9	48.6	45.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.7	44.5	46.7	56.7	64.7	67.0	72.4	71.1	66.2	59.9	54.8	49.4
		MCDB	48.1	45.5	49.8	62.0	70.9	73.9	78.8	76.0	71.0	61.6	55.9	50.3
	2%	WB	44.8	43.2	44.9	54.1	61.1	64.2	69.2	68.2	63.5	58.0	52.6	47.9
		MCDB	45.8	44.1	46.7	58.9	67.2	69.1	75.4	73.7	66.6	59.4	53.9	48.7
	5%	WB	43.6	42.2	43.7	51.8	58.4	62.2	67.1	66.3	61.8	56.6	50.7	46.7
		MCDB	44.7	43.2	45.0	55.9	63.8	66.9	72.7	71.1	64.5	58.0	52.2	47.8
	10%	WB	42.3	41.3	42.6	49.4	55.8	60.1	65.2	64.6	60.3	55.4	49.1	45.4
		MCDB	43.5	42.3	43.8	52.6	60.3	64.0	70.0	68.3	62.9	57.0	50.6	46.7
Mean Daily Temperature Range	5% DB	MDBR	5.4	5.1	5.9	7.8	7.8	6.9	7.3	7.0	6.5	5.5	5.2	5.4
		MCDBR	4.7	4.5	7.2	13.2	14.5	13.3	14.0	11.6	9.2	6.4	4.8	4.4
		MCWBR	5.1	4.1	5.2	8.1	8.7	8.4	7.3	6.3	5.9	5.4	5.4	4.8
	5% WB	MCDBR	4.5	4.3	6.4	12.5	13.3	11.9	12.8	10.3	8.4	5.9	4.7	4.4
		MCWBR	5.2	4.1	5.2	8.1	8.8	8.1	7.4	6.4	6.3	5.6	5.5	4.9
Clear-Sky Solar Irradiance	taub	0.325	0.342	0.362	0.395	0.385	0.382	0.394	0.397	0.377	0.356	0.333	0.309	
	taud	2.229	2.291	2.313	2.269	2.367	2.398	2.379	2.375	2.412	2.384	2.289	2.228	
	Ebn at Noon	177	226	253	260	272	273	267	258	246	219	174	153	
	Edh at Noon	18	25	30	37	36	35	35	33	28	23	17	14	
All-Sky Solar Radiation	RadAvg	145	338	768	1289	1702	1807	1685	1344	908	461	181	104	
	RadStd	25	49	72	143	160	145	172	102	90	54	27	12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)	+0.70	N/A	N/A	N/A	N/A	N/A	-234	-328	N/A	N/A

Nomenclature: See separate page